EPSR-test-ex96 szeros /= 0: extrapolation of the data to Q = 0

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the `szeros` option of the `epsr` engine of EPSR [1,2]: a non-zero value fixes the Q = 0 limit of the interference differential cross section, and the data below the minimum Q are replaced by a linear extrapolation towards this limit, both for the background subtraction and for the fit of the empirical potential [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`szerostemp` = `szeros - sum(ciwt*soqintra(1,:))`, linear extrapolation below `nqmin` in the difference `dts` before `ftback`)
- `epsr_standalone_rev3.f`: `ptgnsq` (extrapolated `qtemp` below max(qminp/qminm, q(nqmin)), `szeros /= 0` R-factor branch), `fit_expansion`

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::iterate` (`szerostemp`, extrapolation); `structure.rs`: `ftback`
- `potential.rs`: `ptgnsq`, `fit_expansion`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. For this case `szeros` is -0.30 for data set 1, -0.12 for data set 2 and 0.0 for data set 3 (mixed case); below `nqmin` the difference is the line conss0 + grads0·(Q - Q(1)) minus the intramolecular term, joining the Q = 0 limit to the first used data point.

Inputs

Only the `.EPSR.inp` differs from EPSR-test, edited by `scripts/make_inp.sh` (via `scripts/setkw.py`): `szeros` -0.30 / -0.12 / 0.0.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
`nqmin` = 8 > 1, so the extrapolation branch is taken. The extrapolated values are visible in the first columns of `delfit.dat` for data sets 1 and 2 and are zero for data set 3; the szeros /= 0 R-factor branch (summing from point 1) only affects the log. All files including `delfit.dat` are identical.

Notes

The `fit_expansion` call is the same in both Fortran branches.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.